

The Contest Toolkit

Angle chasing and hidden circles

Everything here is built from what you already proved. No new rules. Keep these five next to you: (1) a triangle's angles add to 180; (2) equal sides face equal angles, and back again; (3) an exterior angle equals the two far angles added; (4) a rim angle is half the center angle, and every rim point on one arc sees a chord at the same angle; (5) an angle on a diameter is 90, and backwards, anything that sees a segment at 90 sits on the circle with that segment as a diameter.

One habit beats every trick: write every angle you already know onto the picture before you hunt for the one you want.

Warm-ups

W1. Triangle ABC has $AB = AC$ and angle $A = 50^\circ$. (a) Find angles B and C . (b) Split angle B in half with a line hitting AC at D . Find angle ABD and angle ADB .

W2. Four tips of a five-pointed star measure 30° , 25° , 40° , 45° . Find the fifth. (The five tips always add to 180° .)

Angle chasing, the real thing

A1. Triangle ABC has angle $A = 50^\circ$. The lines splitting angle B and angle C in half cross at I . (a) Find angle BIC . (b) Prove that for *any* triangle, angle $BIC = 90^\circ + \frac{1}{2}A$. Notice your answer to (a) never needed B and C on their own.

A2. Triangle ABC has $AB = AC$ and angle $A = 36^\circ$ (so $B = C = 72^\circ$). Split angle B in half, hitting AC at D . Prove $BD = BC$, and then $AD = BD$. (One line makes three equal lengths appear. This triangle builds the regular pentagon.)



Cyclic quadrilaterals and hidden circles

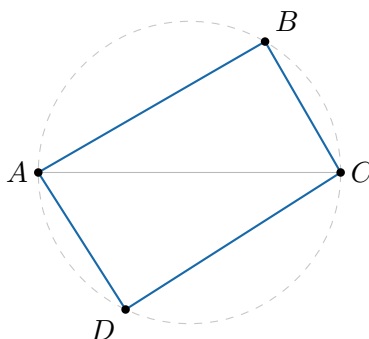
C1. Four points A, B, C, D sit in order on a circle. Prove that opposite angles add to 180° : angle A + angle $C = 180^\circ$. (Each corner is a rim angle; the arcs it stands on cover the whole 360° .)



C2. Points A, B, C, D lie in order around one circle, and angle $CAD = 28^\circ$. (a) Find angle CBD . (b) If also angle $ADB = 35^\circ$, find angle ACB . One sentence each.



C3. Quadrilateral $ABCD$ has right angles at B and D (angle $ABC = \text{angle } ADC = 90^\circ$), shown below. (a) Explain why all four points lie on one circle, and name its diameter. (b) Prove angle $ACB = \text{angle } ADB$.



Constructions (compass and straightedge, no protractor)

D1. Cut a given angle exactly in half. Write the compass-and-straightedge steps, then prove the line you drew really splits the angle evenly. (Look for two triangles with the same three sides.)

D2. Given any triangle, construct the single circle that passes through all three corners. (a) Give the steps. (b) Prove the three perpendicular bisectors of the sides meet at one point, so the construction always works. (Key fact: a point on a segment's perpendicular bisector is the same distance from both ends.)

The challenge

E1. Triangle ABC sits inside its circumcircle (below). Extend the bisector of angle A until it hits the circle again at M . (a) Prove $MB = MC$, so M is the exact middle of arc BC . (b) **Hard.** Let I be the incenter (the crossing of all three angle bisectors). Prove $MI = MB$ too, by showing triangle MBI has two equal angles. Hint: chase angle MBI and angle MIB separately; each comes out to $\frac{1}{2}A + \frac{1}{2}B$.

